

HR Analytics Problem Statement

The organization is experiencing employee attrition across different departments and job roles, impacting productivity, workforce stability, and recruitment costs. HR management requires a comprehensive analytics solution to examine employee demographics, job satisfaction, compensation, work-life balance, overtime, tenure, and performance metrics. The goal is to identify the key drivers of employee turnover, understand workforce trends, detect high-risk employee segments, and develop effective retention strategies to improve employee engagement and organizational performance.

“How can the organization leverage employee data to identify the factors driving attrition, improve employee retention, enhance workforce satisfaction, and support strategic HR decision-making?”

Deliverables

1. **Data Preparation & Modeling (Python):** Cleaned, transformed, and preprocessed HR employee data for analysis.
2. **Data Analysis (MySQL):** Performed SQL queries to analyze employee demographics, attrition, compensation, job satisfaction, and workforce trends.
3. **Visualization & Insights (Power BI):** Developed an interactive HR Analytics dashboard highlighting key KPIs, attrition patterns, workforce distribution, and retention insights.
4. **Report & Recommendations:** Prepared a report summarizing key findings, attrition drivers, and actionable recommendations for employee retention.
5. **GitHub Repository:** Maintained a repository containing Python scripts, SQL queries, Power BI dashboard files, documentation, and project resources for reproducibility and collaboration.